
EXERCISE

Exercises 1

AI510 - Cybersecurity and Innovation

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1 Exercise 1

1.1 Explain, in your own words, the three protection goals that make up the C-I-A triad. Which two protection goal were added later on?

- Confidentiality (data should be unavailable to others)
- Integrity (guarding information)
- Availability (backups)
- Authenticity (ensure that those who are sharing data are who they say they are)
- Non-repudiation. (ensuring that both sender and receiver knows that data sharing has taken place.)

1.2 How was risk mathematically defined in the lecture?

$$R_a = p_a \cdot d_a$$

probability of attack $\times$ damage

1.3 Which questions are indicative of a security mindset?

- Which part of the system can the user control
- What can they do outside of what is intended from the system
- Is any of the unintended functionality useful

1.4 What are the three steps when creating an attack tree?

- Create the tree
- Prune the tree (just gray out things that are already invalid)
- Think about countermeasures for all leafs

1.5 What is the defining property of symmetric encryption opposed to asymmetric encryption?

Symmetric encryption shares 1 key which can be vulnerable, asymmetric encryption uses both public and private key such that Alice can use Bob's public key to encrypt data that only Bob can decrypt.

1.6 What are the three defining properties of cryptographic hash functions?

- One way function (its infeasible to find x s.t. $H(x) = h$ given an h)
- Collision-resistance (No pair (x, y) exists such that $H(x) = H(y)$)
- Resource efficient

1.7 Briefly explain, in your own words, the basic idea behind hybrid encryption.

Hybrid encryption, encrypts a symmetric session k , which can more efficiently decipher the message m . The session k is onetime use.

2 Exercise 2

Assume your goal is to get hold of the written exam for a specific course. You know that the laptop of the respective lecturer is consistently unlocked when they go to the bathroom. However, their office is on a restricted corridor.

2.1 Draw an attack tree for this scenario. Consider all three steps of creating an attack tree separately and draw the tree in all three steps. You don't need a comprehensive tree, but make sure to have at least a few full paths with countermeasures.

1. Wait for the professor to go to the bathroom
 - 1.1. Get through the door to the restricted corridor
 - 1.1.1. steal a key to the corridor
 - 1.1.1.1. steal a key from another employee
 - **countermeasure** track key usage with logs
 - 1.1.2. forge a key to the corridor
 - **countermeasure** Keys should be difficult to forge
 - 1.1.3. break the door to the corridor

- **countermeasure** Alarm
- 1.1.4. ~~teleport into the office~~

2.2 How does the tree change, if the lecturer shares the office with another lecturer? Draw the tree again or add a respective sub-tree in this scenario.

1. Wait for the professor to go to the bathroom
 - 1.1. Get through the door to the restricted corridor
 - 1.1.1. steal a key to the corridor
 - 1.1.1.1. steal a key from another employee
 - 1.1.1.1.1. run past the other professor
 - **countermeasure** expel student
 - **countermeasure** track key usage with logs
 - 1.1.2. forge a key to the corridor
 - **countermeasure** Keys should be difficult to forge
 - 1.1.3. break the door to the corridor
 - **countermeasure** Alarm
 - 1.1.4. ~~teleport into the office~~
2. Wait for both professors to go to the bathroom
 - **countermeasure** professors should not go to the bathroom at the same time.
3. Kill the other professor beforehand
 - **countermeasure** replace the dead professor.

3 Exercise 3

3.1 Explain the term Layered Security in your own words.

One can assume that given enough layers of security. A perpetrator will be caught.

4 Exercise 4

Assume a scenario like the “red telephone”, where two parties need to repeatedly communicate confidential information.

4.1 Which type of encryption is suitable for this scenario? Explain your answer.

Asymmetric encryption (maybe hybrid)

4.2 Assume the number of parties grows to five. Which type of encryption is suitable for this scenario? Explain your answer.

Still asymmetric encryption (maybe hybrid)

5 Exercise 5

5.1 What are the benefits of hybrid encryption?

Mostly efficiency

5.2 Can you imagine a scenario where this is used today? Give an example and explain why it makes sense in this case.

Almost everywhere, for example https

6 Exercise 6

6.1 Encrypt the bitstring $m = 10110110\ 01100111$ using with the key $k = 01010110\ 01011100$ using the One-Time-Pad algorithm. Assume bit-wise encryption.

$$10110110\ 01100111 \oplus 01010110\ 01011100 = 11100000\ 00111011$$